

```
Windows PowerShell
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> notepad app\ingest.py
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> Remove-Item -Recurse -Force chroma_db
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> python -m app.ingest
=====
RAG DOCUMENT INGESTION
=====

Loaded 3 documents.
- ai_overview.txt
- deep_learning.txt
- machine_learning.txt

Created 8 document chunks.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%| 103/103 [00:00<00:00, 1560.43it/s]

ChromaDB vector store created successfully.
Database location: chroma_db
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> notepad app\rag_pipeline.py
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> pip install -U langchain-groq
Requirement already satisfied: langchain-groq in .\venv\Lib\site-packages (1.1.3)
Requirement already satisfied: groq<1.0.0,>=0.30.0 in .\venv\Lib\site-packages (from langchain-groq) (0.37.1)
Requirement already satisfied: langchain-core<2.0.0,>=1.4.0 in .\venv\Lib\site-packages (from langchain-groq) (1.6.6)
Requirement already satisfied: anyio<5,>=3.5.0 in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (4.15.1)
Requirement already satisfied: distro<2,>=1.7.0 in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (0.28.1)
Requirement already satisfied: pydantic<3,>=1.9.0 in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (2.13.5)
Requirement already satisfied: sniffio in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (1.3.1)
Requirement already satisfied: typing-extensions<5,>=4.10 in .\venv\Lib\site-packages (from groq<1.0.0,>=0.30.0->langchain-groq) (4.16.0)
Requirement already satisfied: idna>=2.8 in .\venv\Lib\site-packages (from anyio<5,>=3.5.0->groq<1.0.0,>=0.30.0->langchain-groq) (3.20)
Requirement already satisfied: certifi in .\venv\Lib\site-packages (from httpx<1,>=0.23.0->groq<1.0.0,>=0.30.0->langchain-groq) (2026.7.22)
Requirement already satisfied: httpcore==1.* in .\venv\Lib\site-packages (from httpx<1,>=0.23.0->groq<1.0.0,>=0.30.0->langchain-groq) (1.0.7)
=====
```

```
Windows PowerShell
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> python -m app.chatbot
=====
RAG Q&A CHATBOT
LangChain + ChromaDB + Groq
=====

Ask questions about the indexed documents.
Type 'exit' to quit.

You: What is machine learning?
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%| 103/103 [00:00<00:00, 1713.50it/s]

Assistant:
Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every individual task.

Sources:
- machine_learning.txt
- ai_overview.txt

You: What are the three common categories of machine learning?
Loading weights: 100%| 103/103 [00:00<00:00, 1423.34it/s]

Assistant:
Supervised learning, unsupervised learning, and reinforcement learning.

Sources:
- machine_learning.txt
- ai_overview.txt

You: What are transformers used for?
Loading weights: 100%| 103/103 [00:00<00:00, 1392.42it/s]

Assistant:
Transformers are used primarily for natural language processing tasks and for building modern generative AI systems, leveraging attention mechanisms.
```

```
Windows PowerShell
Loading weights: 100%| 103/103 [00:00<00:00, 1392.42it/s]

Assistant:
Transformers are used primarily for natural language processing tasks and for building modern generative AI systems, leveraging attention mechanisms to model relationships between elements in input sequences.

Sources:
- deep_learning.txt
- ai_overview.txt
- machine_learning.txt

You: What is the population of Mars?
Loading weights: 100%| 103/103 [00:00<00:00, 1301.44it/s]

Assistant:
I could not find enough information in the provided documents to answer this question.

Sources:
- deep_learning.txt
- ai_overview.txt

You: exit

Goodbye!

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> pytest -q

no tests ran in 0.06s
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> dir tests
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> notepad tests\test_rag.py
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> dir tests

Directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma\tests

Mode                LastWriteTime         Length Name
----                -
d-----                06-10-2026 14:14         .pytest_cache
d-----                06-10-2026 13:50         .venv
d-----                06-10-2026 14:05         app
d-----                06-10-2026 14:10         chroma_db
d-----                06-10-2026 13:31         data
d-----                06-10-2026 14:16         tests
-a-----                06-10-2026 13:46          69 .env
-a-----                06-10-2026 13:47          69 .env.example
-a-----                06-10-2026 13:47          79 .gitignore
-a-----                06-10-2026 14:16          24 pytest.ini
-a-----                06-10-2026 14:17        5706 README.md
```

```
Windows PowerShell

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> pytest -q
.... [100%]
4 passed in 2.59s
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> notepad README.md
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> dir

Directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma

Mode                LastWriteTime         Length Name
----                -
d-----                06-10-2026 14:14         .pytest_cache
d-----                06-10-2026 13:50         .venv
d-----                06-10-2026 14:05         app
d-----                06-10-2026 14:10         chroma_db
d-----                06-10-2026 13:31         data
d-----                06-10-2026 14:16         tests
-a-----                06-10-2026 13:46          69 .env
-a-----                06-10-2026 13:47          69 .env.example
-a-----                06-10-2026 13:47          79 .gitignore
-a-----                06-10-2026 14:16          24 pytest.ini
-a-----                06-10-2026 14:17        5706 README.md

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> dir app

Directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma\app

Mode                LastWriteTime         Length Name
----                -
d-----                06-10-2026 14:09         __pycache__
-a-----                06-10-2026 14:05        1042 chatbot.py
-a-----                06-10-2026 14:09         383 config.py
-a-----                06-10-2026 14:02        2276 ingest.py
```

```
Windows PowerShell X + -
```

```
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> pytest -q  
..... [100%]  
4 passed in 2.41s  
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma> python -m app.chatbot  
=====  
RAG Q&A CHATBOT  
LangChain + ChromaDB + Groq  
=====
```

```
Ask questions about the indexed documents.  
Type 'exit' to quit.
```

```
You: What is machine learning?  
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downlo  
ads.  
Loading weights: 100%|██████████████████████████████████████| 103/103 [00:00<00:00, 1215.58it/s]
```

```
Assistant:  
Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or de  
cisions without being explicitly programmed for every individual task.
```

```
Sources:  
- machine_learning.txt  
- ai_overview.txt
```

```
You: What are the three common categories of machine learning?  
Loading weights: 100%|██████████████████████████████████████| 103/103 [00:00<00:00, 1613.19it/s]
```

```
Assistant:  
Supervised learning, unsupervised learning, and reinforcement learning.
```

```
Sources:  
- machine_learning.txt  
- ai_overview.txt
```

```
29°  
 14:53  
06-10-2026
```

```
Windows PowerShell X + v
You: What is machine learning?
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|██████████████████████████████████████| 103/103 [00:00<00:00, 1215.58it/s]

Assistant:
Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every individual task.

Sources:
- machine_learning.txt
- ai_overview.txt

You: What are the three common categories of machine learning?
Loading weights: 100%|██████████████████████████████████████| 103/103 [00:00<00:00, 1613.19it/s]

Assistant:
Supervised learning, unsupervised learning, and reinforcement learning.

Sources:
- machine_learning.txt
- ai_overview.txt

You: What is the population of Mars?
Loading weights: 100%|██████████████████████████████████████| 103/103 [00:00<00:00, 1629.61it/s]

Assistant:
I could not find enough information in the provided documents to answer this question.

Sources:
- deep_learning.txt
- ai_overview.txt

You: exit

Goodbye!
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\RAG_LangChain_Chroma>
```